

PAPER 1 / QUESTION 6

8820-6501 / original question 6

6. An object of mass $2m$ moving at velocity $3v$ collides with a stationary object of mass $4m$. The objects stick together after the collision. What is the final speed and the change in total kinetic energy immediately after the collision?

	Final speed	Change in total kinetic energy
A.	v	$3 mv^2$
B.	v	$6 mv^2$
C.	$2v$	$3 mv^2$
D.	$2v$	$6 mv^2$